

LIBRARY BOOK BORROWING SYSTEM

End of semester dynamic web application project- option E

GROUP NAMES

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SYSTEM ADMINISTRATION

The library Book Borrowing System is a PHP and MySQL web application for managing student book loans. Students can register, login, search the catalogue by title or author, borrow available books, return books, and view their borrowing history. Admin users can manage book records and monitor borrowing activity.

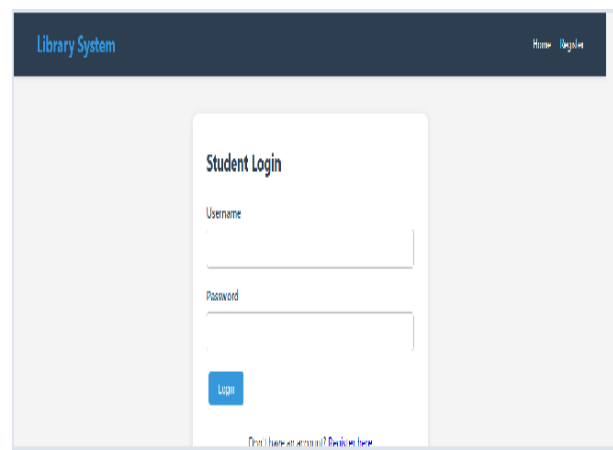
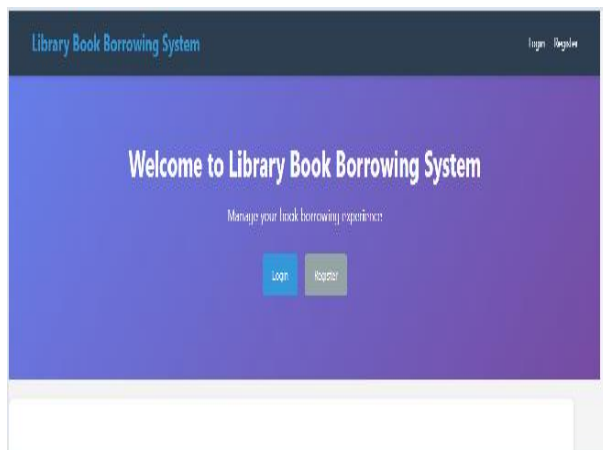
Functional requirements covered

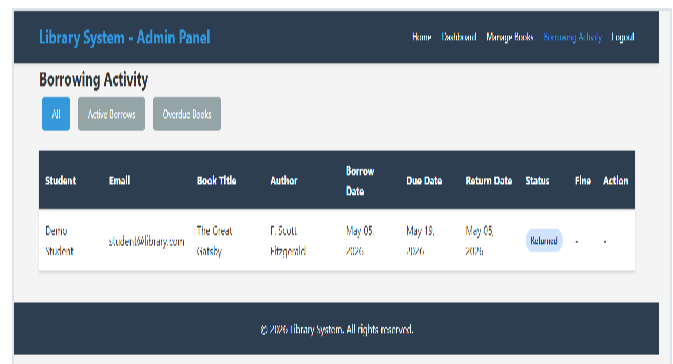
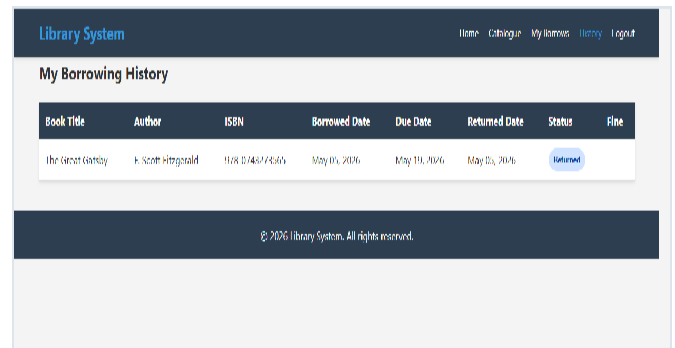
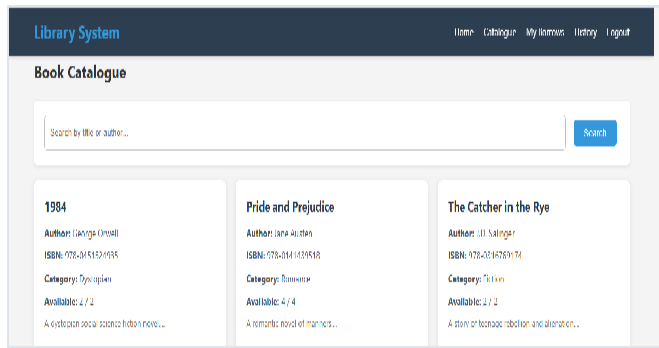
Requirements	Implementation
Registration, login, logout, and sessions	'User.php', 'register.php', 'login.php' and 'logout.php' manage authentication and protected pages.
Forms, validation, and persistent data	Student registration, book borrowing, returns and admin book forms store data in MySQL with feedback
Dynamic display, search, and filtering	Catalogue, borrowing history, admin dashboard, and borrowing activity pages retrieve records from MySQL.
Edit, delete, and role-based access	Admins can add, edit, and delete books. Admin pages require the 'admin' role

DATABASE STRUCTURE

TABLE	MAIN COLUMNS	PURPOSE
Users	User_id, username, email, password, full_name, enrollment_id, phone, address, is_active	Stores student and admin accounts
books	book_id, title, author, isbn, publisher, publication_year, category total_copies, available_copies, description	Stores book catalogue records and availability
Borrowing_records	Record_id, user_id, book_id, borrow_date, return_date, status, fine_amount	Stores borrowing and return transactions.

KEY SCREENS





Challenges and lessons learned

One of the key challenges was coordinating borrowing records with a book availability so that returned books become available again and active loans are protected from accidental deletion.

Another challenge was keeping a student and admin permissions separate while allowing both roles to use the system smoothly.

Meeting up as a group was also tough, time management was also a problem.

Lessons learnt

How to connect PHP pages to MySQL database using prepared statements, manage user sessions, validate form input, and to organize a dynamic web application into a reusable classes for users, books, borrowing records, and database access.